AI CLASS · PROJECT PROPOSAL · SPEAKING SCRIPT

# AI Model *Lab*

*Click it. Test it. See the result.*

This is a full script, one part per presenter. Read it a few times, then say it in your own words: it sounds more natural than reading word for word. Grey lines with ▸ are stage directions, and `SPACE` means press Space (or the clicker) to go to the next slide.

| PART | PRESENTER | SLIDES | TIME |
|---|---|---|---|
| 1 · The lab | A | Intro video, 1, 2 | about 2:45 |
| 2 · The making | B | 3, 4 | about 3:30 |
| 3 · The models | C | 5, 6 | about 2:45 |
| 4 · What's next | D | 7, 8 | about 2:00 |
| 5 · Conclusion | E | 9 (then questions) | about 1:15 |
| **Total** | | | **about 12 minutes** |

Times are at a calm speaking pace (about 130 words a minute). If your slot is shorter, skip the optional ▸ banner notes first, then shorten slides 3 and 4.

**Before you start:** open `model-lab-library.vercel.app/present` on the classroom computer, wait for "Ready" under the start button, turn the sound on, and press **F** for full screen. Replace "[name]" in the script with each presenter's name.

**One rule:** every number in this script is measured. Don't add numbers that aren't here. If the professor asks about something we haven't measured, say: "We haven't measured that yet."

PART 1 · THE LAB

Presenter A

# Intro video, slides 1 and 2

## Before the video

about 0:40 with the video

Good morning, everyone. Good morning, Professor. We are [group name], and today we're presenting our project: AI Model Lab. Let's start with a short video.

- *Press Space to start. Let the 21-second video play without talking. It stops on its last frame.*

SPACE

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## Slide 1 · Topic and team

about 0:55

So, what is AI Model Lab? It's a library of small AI models that anyone can try, right in a web browser. Our motto is on the screen: **click it, test it, see the result.**

Each model in the lab does one job, and does it well. And each one comes with an honest report card. Not just a score, but also the cases where the model gets it wrong.

There are five of us. I'm [name A], and I'll introduce the lab. [Name B] will explain how we made the models. [Name C] will show you the two working models and their results. [Name D] will talk about what we're working on now, and [Name E] will wrap up.

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## Slide 2 · What the lab does

about 1:10

Here's what the lab can do.

First, you can try a small AI model right in your browser. There's nothing to install.

Second, you can test it in different ways: use one of our sample photos or sounds, upload your own, or let it listen live through the microphone.

Third, every model has an honest report card. It shows the score we measured, and it shows how the model fails.

Fourth, nothing you add is uploaded. The model downloads to your phone or laptop and runs there, so your photo or your recording never leaves your device.

And right now, **two models are live**: a sea creature detector and a sound detective. More are planned.

Now [name B] will explain how we actually built these models.

SPACE

PART 2 · THE MAKING

Presenter B

# Slides 3 and 4

### Slide 3 · How the models learn

about 1:45

Thank you, [name A]. I'll explain the AI techniques we used.

The first one is **supervised learning**. The model learns from examples where we already know the right answer. For the sea creature detector, we used **7,637 photos** from Open Images and the Aquarium Combined dataset, with every creature marked by a box. For the sound detective, we used **13,825 sound clips** from FSD50K and Freesound, each labelled with the sounds it contains.

The second technique is **transfer learning**. We didn't train from zero. We started from models that were already trained on huge datasets, and fine-tuned them on our data. For photos, that's YOLO26 Nano, a small object-detection network. For sound, that's EfficientAT, which was trained on AudioSet.

Third, we split the data fairly into training, validation and test sets. The test set stays frozen: **767 photos and 4,216 sound clips** that the models never see during training. So our scores come from data the models have never seen.

And finally, sound is **multi-label**. We turn the audio into a spectrogram, which is basically a picture of sound. The model looks at one-second slices, every half second, and one slice can contain several sounds at the same time.

▸ *If there's time, point at the banner: the photos move through Collect, Train, Test and Ship, and at Test each one gets the box our real model gave it.*

SPACE

### Slide 4 · How they predict, and how we measure

about 1:45

Next: how the models make predictions, and how we measure them.

The sea model does **object detection**. It draws a box around each creature and gives a confidence score. It often draws several boxes on the same animal, so we use **IoU**, which measures how much two boxes overlap, and **non-max suppression**, which keeps only the best box.

Then we use **thresholds**. A creature is shown only if the model is at least 45% sure. The sound model has a separate threshold for each of its 17 sounds. We tuned those on the validation set, never on the test set, so the test score stays honest.

To measure the models, we use **mAP**, mean average precision. For the detector we use mAP at 50, which means a box counts as correct if it overlaps the real box by at least half. For sound we also use **F1**. And for each model we keep a list of how it fails.

Last, both models are exported to **ONNX** and run with ONNX Runtime Web inside the browser. That's why nothing gets uploaded.

Now [name C] will show you the two models working, and their real results.

`SPACE`

PART 3 · THE MODELS

Presenter C

# Slides 5 and 6

## Slide 5 · Sea creature detector

about 1:25

Thank you, [name B]. This part of the proposal asks for the expected result as a UI prototype. But as you can see, the label just changed from "expected" to **"built"**. This is already working, and the video on the right is a real screen capture of the live page.

This is our first model, the **sea creature detector**. You give it a photo, and it finds and boxes 14 kinds of sea creatures: fish, sharks, dolphins, sea turtles, penguins and more.

On 767 test photos it never saw, it scores **0.74 mAP at 50**. On aquarium photos, it does even better: **0.82**.

And here's the honest part. It mixes up seals and sea lions, and dolphins and whales, and it misses small fish that are far away. You can see real mistakes in the video: it calls a seal a sea lion, it calls a shark a plain fish, and in the school of fish it finds only 9 of the 23 fish.

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## Slide 6 · Sound detective

about 1:20

Our second model is the **sound detective**. It listens for 17 everyday sounds, like a dog barking, a siren, glass breaking or knocking, and shows when each one happens on a timeline. You can record, upload a file, or let it listen live.

The video shows its real output for one of our samples, called "Two sounds at once". You can see it hearing speech, traffic and clapping at the same time, each with its confidence.

On 4,216 test clips, it scores **0.79 mean average precision**. We also made a design choice here: a bigger version scores 0.79 at 20.5 megabytes, and a smaller one scores 0.76 at 6.5 megabytes. We chose the more accurate one.

Its weak spots: the doorbell is its hardest sound, footsteps and car horns are also difficult, and the timing is still rough. It finds about half of the sounds within half a second of when they really start.

Now [name D] will tell you what we're working on next.

SPACE

PART 4 · WHAT'S NEXT

Presenter D

# Slides 7 and 8

## Slide 7 · RAG assistants

about 1:10

Thank you, [name C]. Besides the two models, we're also building something bigger: **RAG assistants**.

RAG means retrieval-augmented generation. When you ask a question, the assistant first searches its own documents for the passages that answer it. Then a small language model writes the reply using only those passages. So it answers from its documents, not from guessing.

Like everything else in the lab, it runs fully in the browser. It uses Qwen 2.5, a 1.5-billion-parameter language model, plus a small search model, on WebLLM.

We've built two assistants: one for a university, the University of Raffel Luo, and one for a district emergency office, Pathum Rai District.

We've tested them, but the work isn't complete yet. Right now we're measuring how well different defences protect them against tricky prompts, so we don't have final results to share today.

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## Slide 8 · What's coming

about 0:50

So, what's coming next?

First, we'll finish the **RAG testing**: we'll measure each defence on its own, on both assistants, and then all of them combined. After that, a private document assistant will be our next test target.

For the sea detector, we want to add the creatures it can't find yet, like octopus, manta ray, orca and lionfish. For the sound detective, a version 2 with better timing and more sounds.

And later, a speech translator and text models.

Now [name E] will wrap up our presentation.

SPACE

PART 5 · CONCLUSION

Presenter E

# Slide 9 and questions

### Slide 9 · What we built

about 1:15

Thank you, [name D]. To sum up what we built:

We have **two working AI models**, the sea creature detector and the sound detective, and both run on your own device. We have also built **two RAG assistants**, which are still being tested.

What makes our lab different is honesty. Every number we showed you today is measured, and every model shows how it fails, not just how well it does.

Our references are on the screen. The data comes from Open Images, the Aquarium Combined dataset, FSD50K and Freesound. The models are YOLO26, EfficientAT, Qwen 2.5 and snowflake-arctic-embed. We used ONNX Runtime Web, WebLLM and Remotion for the videos, and the music is by Alex Morgan from Pixabay.

Everything is live right now. Please scan the QR code and try the models on your phone.

Thank you for listening. We're happy to answer your questions.

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### If you get questions

anyone can answer

**"Why run the models in the browser?"**
Privacy and cost: the photo or sound never leaves the device, and we don't need a server to run the models.

**"Why such small models?"**
They have to download quickly and run on a phone. That's also why we compared the 20.5 MB and 6.5 MB sound models.

**"How do you know the scores are fair?"**
They come from a frozen test set the models never saw during training, and the thresholds were tuned on the validation set, not the test set.

**"What are the RAG results?"**
"We're still measuring the defences, so we don't have final results yet. We'll share them once the testing is complete."

**Anything we haven't measured:**
"We haven't measured that yet."